Reading group notes: retrieval-augmented generation

Reading group of the Ashbury machine learning lab

1 Background

Practitioners report that the chunk size is the first component to fail on long inputs. A practical system has to balance the cost of the dense embedding against the quality of the citation. Recent work has made the retriever a central concern, because it decides how far retrieval-augmented generation models can scale. Despite steady progress, the interaction between the citation and the query rewrite is still poorly understood.

The context window also makes retrieval-augmented generation easier to inspect, since its behaviour can be read off per token. We expect the query rewrite to interact with the chunk size at larger scales, which we leave to future work. A limitation of retrieval-augmented generation is that the reranker must be tuned separately for each model size. Combining retrieval-augmented generation with methods that target the citation is a promising direction.

A practical system has to balance the cost of the grounding check against the quality of the context window. Despite steady progress, the interaction between the query rewrite and the reranker is still poorly understood. Recent work has made the reranker a central concern, because it decides how far retrieval-augmented generation models can scale. Most prior methods treat the context window as fixed and tune the query rewrite around it.

2 Methods we read

To keep training stable, retrieval-augmented generation normalises the query rewrite before it is combined with the retriever. Each block of retrieval-augmented generation applies the citation after a short convolution of width 8. A coefficient controls how strongly the grounding check depends on the reranker; we set it to 0.8 throughout. We implement the grounding check as a fused kernel, which avoids writing intermediate results to memory.

The chunk size in retrieval-augmented generation is parameterised by a small linear layer with 512 hidden units. A coefficient controls how strongly the reranker depends on the dense embedding; we set it to 0.6 throughout. The query rewrite is initialised so that retrieval-augmented generation starts training close to BM25. Retrieval-augmented generation computes the keyword index once per layer and reuses it for every head.

To keep training stable, retrieval-augmented generation normalises the context window before it is combined with the passage index. A coefficient controls how strongly the passage index depends on the citation; we set it to 0.7 throughout. Unlike BM25, retrieval-augmented generation updates the chunk size only when the query rewrite changes. Each block of retrieval-augmented generation applies the keyword index after a short convolution of width 2.

3 Results we discussed

Variance across seeds stays below 0.9 points for every configuration we report. The gains of retrieval-augmented generation grow with the input length, as the reranker becomes the bottleneck. Retrieval-augmented generation scores 44.2% on SciDocs-Lite, compared with 73.5% for a dense retriever. On CiteBench, retrieval-augmented generation improves over BM25 by 1.1 points.

Retrieval-augmented generation scores 57.4% on CiteBench, compared with 59.2% for a dense retriever. Variance across seeds stays below 0.7 points for every configuration we report. Throughput is measured with a batch size of 512 on a single accelerator. We train every model for 30B tokens with the same data order and report the mean of three seeds.

Retrieval-augmented generation scores 75.4% on CiteBench, compared with 42.0% for BM25. We train every model for 80B tokens with the same data order and report the mean of three seeds. On SciDocs-Lite, retrieval-augmented generation improves over a closed-book model by 6.6 points. With the passage index frozen, retrieval-augmented generation still recovers most of its improvement on SciDocs-Lite.

4 Open questions

We expect the reranker to interact with the query rewrite at larger scales, which we leave to future work. The passage index also makes retrieval-augmented generation easier to inspect, since its behaviour can be read off per token. A limitation of retrieval-augmented generation is that the retriever must be tuned separately for each model size. Combining retrieval-augmented generation with methods that target the dense embedding is a promising direction.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The query rewrite also makes retrieval-augmented generation easier to inspect, since its behaviour can be read off per token. Combining retrieval-augmented generation with methods that target the grounding check is a promising direction. We expect the citation to interact with the chunk size at larger scales, which we leave to future work.